

Assignment 1 - Theory of Dynamic Systems

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Evaluation is mostly based on how clearly you explain concepts and motivate statements, rather than computations. If you show the correct procedure but you make a mistake in the computation, it won't affect how the exercise is evaluated, as long as the procedure is properly motivated.

Exercise 1 (25 Points)

Given the dynamical matrix

$$A = \begin{bmatrix} 3 & -1 & 0 \\ 0 & -4 & 2 \\ 0 & 0 & -1 \end{bmatrix}$$

compute the eigenvectors, eigenvalues, and assess the stability of the dynamical system $\dot{x} = Ax$ in free evolution.

Exercise 2 (25 Points)

Given the dynamical matrix

$$A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

compute the eigenvectors, eigenvalues, and assess the stability of the dynamical system $\dot{x} = Ax$ in free evolution.

Exercise 3 (40 Points)

Consider the dynamical matrix of Exercise 1, and the system

$$\begin{aligned}\dot{x} &= Ax + Bu \\ y &= Cx\end{aligned}$$

where

$$B = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 0 & 1 \end{bmatrix}.$$

Compute the input/output transfer function $W(s)$, and motivate its shape with the controllability and observability tests. Do the same using the same A and B , and $C = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}$.

Exercise 4 (10 Points)

Fill in the python script `hw01.py`. You have to plot the free evolution of the system $\dot{x} = Ax$, where the matrix A is the one specified in exercise 1. Test the evolution with different initial conditions. What behavior do you expect, given the stability analysis performed in exercise 1?